NDEYE MAGUETTE MBAYE - PhD

Research Scientist - Applied AI & LLMs | Healthcare

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About Me

Research Scientist with a PhD in Machine Learning, specialized in multimodal and LLM-based models for healthcare applications.
I have experience building end-to-end pipelines, from data curation to model training, and evaluation on real-world medical datasets. I also care deeply about robustness and interpretability of AI systems deployed in sensitive settings such as medical ones.

Selected Work Experiences

Ph.D. Student - Machine Learning & Multimodal Modeling **September 2019 - September 2024 | 5 years**

Center for Computation Biology | Institut Curie, Mines Paris PSL and INSERM | Paris, France.

- Developed classical ML models and Transformer-based models for multimodal Electronic Health Records (text + tabular + sequential) to predict breast cancer prognostic status
- Designed late fusion Transformer architectures (based on BEHRT), improving predictive performance (AUC +10%)
- Ran comparative experiments across 3 embedding strategies (Time embeddings using time2vec, Numerical value embeddings for continuous features, contextual embeddings), improving AUC from 0.68 to 0.75 on downstream BC prognostic prediction
- Analyzed model behavior and interpretability to support trust in medical decision-making
- Collaborated with domain experts (clinicians / medical researchers) to interpret results and validate findings
- Co-authored a paper, presented results in 5+ international conferences
- Built reproducible and scalable pipelines for real-world medical datasets

Deep Learning intern **November 2022 - January 2023 | 3 months**

Max Planck Institute for Psychiatry | Munich, Germany

- Developed Tansformer-based models for multimodal EHR using multiple integration methods
- Benchmarked 4+ embedding strategies for patient trajectory modeling, improving prediction performance by ≈ 7%

Deep Learning intern **June - September 2021 | 3 months**

IBM Research Lab | Haifa, Israel

- Data mining large scale datasets (MIMIC II databases, Institut Curie's datasets)
- Explored Tansformer-based models for tabular EHR based on the BEHRT model

Bioinformatics research assistant **April - December 2018 | 8 months**

Joint Microbiology Laboratory | Development Research Institute (DRI), Senegalese Institute for Agricultural Research (SIAR) and Cheikh Anta DIOP University of Dakar (CADU) | Bel Air, Dakar

- Developed dynamic boolean model of gene regulatory network involved in root development in *Arabidopsis thaliana*.
- Validated built model with other species
- Informed on critical genes involved in lateral roots development and highlighted the differential expressions of genes in that process.

Applied AI / LLM Projects

- HospiNet – EUvsVirus Hackathon: Developed a healthcare-focused ML solution to optimize hospital resource allocation during crises.
- Resume Screening Assistant: Built an assistant that analyzes CVs to extract key information, generate summaries, and answer structured questions, with careful evaluation of result quality and reliability.
- Conversational Chatbot: Developed a personal assistant chatbot to help users navigate documents, answer questions, and summarize content.

Education

Ph.D. in Bioinformatics | AI for cancer prognosis, Mines Paris PSL, France 2019 – 2024

Master in Biomathematics – Bioinformatics, Cheikh Anta DIOP University of Dakar, Senegal 2016 – 2019

Bachelor in Biomathematics – Bioinformatics, Cheikh Anta DIOP University of Dakar, Senegal 2015 – 2016

Bachelor in Biology, Chemistry and Geosciences, Cheikh Anta DIOP University of Dakar, Senegal 2012 – 2015

Technical Skills

Core Expertise	Large Language Models (pretraining, representation learning, fine-tuning, prompting, evaluation etc.) Multimodal ML (text, structured, temporal data) Retrieval-Augmented Generation and knowledge injection Evaluation frameworks
AI Tools	Machine Learning libraries (numpy, pandas, scikit-learn etc.), Deep Learning frameworks (Pytorch, Pytorch Lightning), Generative/Agentic AI frameworks (Huggingface, Transformers, spaCy, BeautifulSoup, NLTK, Langchain, Langgraph etc.), Data visualization (matplotlib, seaborn, ggplot2), MLOps (Weights & Biases, Git), XAI (Shap, LIME, Captum), APIs & Prototyping (Streamlit, FastAPI), Accelerated computing (CUDA, SLURM).
Programming Languages	Python, R, Bash, SQL.

Professional Skills

Communication	Lead writer of scientific articles Speaker in international conferences Lead author of AI findings reports to cross-functional teams
Project management	Owner of end-to-end research ML/NLP projects Managed technical scope, timelines, and collaborations
Research culture	Proactive learner, adaptability to new challenges and environments, problem-solving mindset, autonomy
Languages	French : Bilingual English : C1

Selected Talks at Conferences

- **Women in Machine Learning and Data Science**, more information here | Paris, France | September 20th, 2022.
- **Machine Learning Frontiers for Precision Medecine**, more information here | Munich, Germany | October 18th, 2022.
- **Winter School in Computer Science and Engineering**, more information here | Jerusalem, Israël | January 15th, 2023.
- **Computational Systems Biology of Cancer**, more information here | Paris, France | September 25th, 2023 – ***with the Award of the best talk in Ph.D. Students category.***
- **Personalized Health Conference**, more information here | Basel, Switzerland | March 3rd, 2024.

Scientific Publication

- Mbaye NM, Danziger MM, Rosen-Zvi M, Toussaint A, Dumas E, Guerin J, Hamy-Petit A-S, Reyal F and Azencott C-A (2025) Multimodal BEHRT: transformers for multimodal electronic health records to predict breast cancer prognosis. Front. Oncol. 15:1496215. doi: 10.3389/fonc.2025.1496215

Certificates

- **Post-training of LLMs –** *DeepLearning.ai*
- **Langchain and Langgraph Specialization -** *Coursera*
- **Cloud & Deployment fundamentals (AWS) -** *Coursera*
- **Web & API development fundamentals -** *Coursera*

Hobbies

- Cooking
- Football
- Seewing
- Art